Statement 1: Standard Deviation is independent of any change of origin but is dependent on the change of scale**.**

Proof: Let $x_1, x_2, \ldots, x_n$ be a set of observations.

Let the origin of x is changed to h and scale is changed to k.

Then, $y_i = \frac{x_i - h}{k}$

$$\Rightarrow x_i = ky_i + h$$

$$\therefore \bar{x} = h + k\bar{y}$$

$$\therefore x_i - \bar{x} = k(y_i - \bar{y})$$

$$\therefore \sigma_x^2 = \sum \frac{(x_i - \bar{x})^2}{n} = \sum \frac{\left\{k(y_i - \bar{y})^2\right\}}{n} = k^2 \sum \frac{(y_i - \bar{y})^2}{n} = k^2 \sigma_y^2$$

$$\therefore \sigma_x = k\sigma_y$$

Hence the proof.

Statement 2: Central moment are invariant under change of origin but not under scale change.

Proof: Let $x_1, x_2, \ldots, x_n$ be a set of observations.

Let the origin of x is changed to h and scale is changed to k.

Then, $y_i = \frac{x_i - h}{k}$

$$\Rightarrow x_i = ky_i + h$$

$$\therefore \bar{x} = h + k\bar{y}$$

$$\therefore x_i - \bar{x} = k(y_i - \bar{y})$$

$$\therefore \mu_r = \frac{1}{n}\sum_i \left(x_i - \bar{x}\right)^r = \frac{1}{n}\sum_i \left\{k\left(y_i - \bar{y}\right)\right\}^r = \frac{k^r}{n}\sum_i \left(y_i - \bar{y}\right)^r = k^r \mu_{y_r}$$

i.e., $\mu_r = k^r \mu_{y_r}$

Hence the proof.

Example 1: In a test result, number of students in each marks group was recorded. Calculate the Standard Deviation of marks which has the distribution as shown in the table below:

Marks	**Number of Students**
20-29	5
30-39	12
40-49	15
50-59	20
60-69	18
70-79	10
80-89	6
90-99	4

Solution:

Marks	f	Midvalue	$y=\frac{x-54.5}{10}$	fy	fy^2
20-29	5	24.5	-3	-15	45
30-39	12	34.5	-2	-24	48
40-49	15	44.5	-1	-15	15
50-59	20	54.5	0	0	0
60-69	18	64.5	1	18	18
70-79	10	74.5	2	20	40
80-89	6	84.5	3	18	54
90-99	4	94.5	4	16	64
	N=90			$\sum fy=18$	$\sum fy^2=284$

Now, $y=\frac{x-h}{k}$

$$\Rightarrow \sigma_x = k\sigma_y$$

Here, h=54.5, k=10

$$\therefore \sigma_y^2 = \sum\frac{fy^2}{N} - \left(\sum\frac{fy}{N}\right)^2 = \frac{284}{90} - \left(\frac{18}{90}\right)^2 = 3.116$$

$$\therefore \sigma_y = \sqrt{3.116} = 1.77$$

$$\therefore \sigma_x = 10\times1.77 = 17.7$$

Therefore, standard deviation of marks for this distribution is 17.7.

Example 2: Calculate Mean, Median, Mode, Q_1 and D_4 from the following data:

Class	f
0-5	12
5-10	30
10-15	51
15-20	84
20-25	66
25-30	50
30-35	7

Solution:

Class	f	Midvalue	$y=\frac{x-17.5}{5}$	fy	cf
0-5	12	2.5	-3	-36	12
5-10	30	7.5	-2	-60	42
10-15	51	12.5	-1	-51	93
15-20	84	17.5	0	0	177
20-25	66	22.5	1	66	243
25-30	50	27.5	2	100	293
30-35	7	32.5	3	21	300
	N=300			$\sum fy=40$	

Finding Mean:

We know that, Mean, $\bar{x}=h+k\bar{y}$

Here h=17.5 and k=5.

Hence, $\bar{x}=17.5+5\bar{y}=17.5+5\times\frac{\sum fy}{300}=17.5+5\times\frac{40}{300}=18.1667$

$\therefore$ **Mean** =18.1667

Finding Median:

The formula for computing median is.

Median= $l+\frac{\frac{N}{2}-F}{f_m}\times k$

Where,

$N\rightarrow$Total frequency

$l \rightarrow$ lower class boundary of median class

$F \rightarrow cf$ of the immediate preceeding class of median class

$f_m \rightarrow$ frequency of median class

$k \rightarrow$ width of median class

Here, $N = 300, l = 15, F = 93, f_m = 84$ and $k = 5$

$\therefore$ **Median** = $15 + \frac{150 - 93}{84} \times 6 = 18.3928$

<u>Finding Mode:</u>

The formula for computing Mode is,

Mode= $l + \frac{f_k - f_{k-1}}{2f_k - f_{k-1} - f_{k+1}} \times c$

Where,

$l \rightarrow$ lower class boundary of mode class

$f_k \rightarrow$ frequency of the mode class

$f_{k-1} \rightarrow$ frequency of the preceding class

$f_{k+1} \rightarrow$ frequency of the succeeding class

$c \rightarrow$ width of mode class

In the above table the maximum frequency is 84. Therefore, the mode class is 15-20.

Hence, $l = 15, f_k = 84, f_{k-1} = 51, f_{k+1} = 66$ and $c = 5$

$\therefore$ **Mode**= $15 + \frac{84 - 51}{2 \times 84 - 51 - 66} \times 5 = 18.2353$

<u>Finding Quartiles:</u>

$$Q_i = l + \frac{\frac{iN}{4} - F}{f_q} \times c$$

Where,

$l \rightarrow$ lower class boundary of Q_i^{th} class

$F \rightarrow cf$ of the class preceding the Q_i^{th} class

$f_q \rightarrow$ frequency of the Q_i^{th} class

$c \rightarrow$ width of the Q_i^{th} class

Now, Q_1 class: Class with $\left(\frac{n}{4}\right)^{th}$ value of the observation in the cf column

$= \left(\frac{300}{4}\right)^{th}$ value of the observation in the cf column

$= (75)^{th}$ value of the observation in the cf column

Which lies in the class 10-15.

$\therefore Q_1$ class : 10-15

Therefore, $l = 10, F = 42, f_q = 51$ and $c = 5$

$$\therefore Q_1 = 10 + \frac{\frac{1 \times 300}{4} - 42}{51} \times 5 = 13.2353$$

<u>Finding Deciles:</u>

$$D_j = l + \frac{\frac{jN}{10} - F}{f_d} \times c$$

Where,

$l \rightarrow$ lower class boundary of D_j^{th} class

$F \rightarrow cf$ of the class preceding the D_j^{th} class

$f_d \rightarrow$ frequency of the D_j^{th} class

$c \rightarrow$ width of the D_j^{th} class

Now, D_4 class: Class with $\left(\frac{4n}{10}\right)^{th}$ value of the observation in the cf column

$= \left(\frac{4 \times 300}{10}\right)^{th}$ value of the observation in the cf column

$= (120)^{th}$ value of the observation in the cf column

Which lies in the class 15-20.

$\therefore D_4$ class : 15-20

Therefore, $l = 15, F = 93, f_d = 84$ and $c = 5$

$$\therefore D_4 = 15 + \frac{\frac{4\times 300}{10} - 93}{84} \times 5 = 16.6071$$

Example 3: Find the second, third and fourth central moment and hence find skewness (γ_1) and kurtosis (γ_2) from the following data:

Class Limit	f
110.0-114.9	15
115.0-119.9	15
120.0-124.9	20
125.0-129.9	35
130.0-134.9	10
135.0-139.9	10
140.0-144.9	5

Solution:

Class Limit	f	Midvalue(M)
110.0-114.9	15	112.45
115.0-119.9	15	117.45
120.0-124.9	20	122.45
125.0-129.9	35	127.45
130.0-134.9	10	132.45
135.0-139.9	10	137.45
140.0-144.9	5	142.45
	N=110	

Now in order to find the Central moments, we will first find the raw moments.

For raw moments, $\mu_r' = \dfrac{\sum f(X-A)^r}{N}$. We take A=0 here i.e., raw moment about the origin.

fM	fM^2	fM^3	fM^4
1686.75	189675.0375	21328957.97	2398441323
1761.75	206917.5375	24302464.78	2854324488
2449	299880.05	36720312.12	4496402219
4460.75	568522.5875	72458203.78	9234798071
1324.5	175430.025	23235706.81	3077569367
1374.5	188925.025	25967744.69	3569266507
712.25	101460.0125	14452978.78	2058826827
$\sum fM = 13769.5$	$\sum fM^2 = 1730810.275$	$\sum fM^3 = 218466368.9$	$\sum fM^4 = 27689628800$

$$\therefore \mu_1^{'} = \frac{13769.5}{110} = 125.1773$$

$$\mu_2^{'} = \frac{1730810.275}{110} = 15734.6389$$

$$\mu_3^{'} = \frac{218466368.9}{110} = 1986057.899$$

$$\mu_4^{'} = \frac{27689628800}{110} = 251723898.2$$

These raw moments will be used to find central moments $\mu_1, \mu_2, \mu_3, \mu_4$

$$\mu_1 = 0$$

$$\mu_2 = \mu_2^{'} - (\mu_1^{'})^2 = 65.2823$$

$$\mu_3 = \mu_3^{'} - 3\mu_2^{'}\mu_1^{'} + 2(\mu_1^{'})^3 = 94.5197$$

$$\mu_4 = \mu_4^{'} - 4\mu_3^{'}\mu_1^{'} + 6\mu_2^{'}\left(\mu_1^{'}\right)^2 - 3\left(\mu_1^{'}\right)^4 = 10234.9776$$

Therefore, $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{94.5197^2}{65.2823^3} = 0.0321$ and $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{10234.9776}{65.2823^2} = 2.4016$

Consequently, **Skewness** $\gamma_1 = \sqrt{\beta_1} = 0.1792$

And **kurtosis** $\gamma_2 = \beta_2 - 3 = -0.5984$

Hence, the distribution is **positively skewed and platykurtic.**

Example 4: For a group of 200 candidates, the mean and standard deviation of scores were found to be 40 and 15 respectively. Later it was discovered that the scores 43 and 35 were misread as 34 and 53 respectively. Find the corrected mean and s.d. corresponding to corrected values.

Solution: let x be the give variable.

$$n = 200, \bar{x} = 40, \sigma = 15$$

Now, $\bar{x} = \frac{1}{n}\sum_i x_i$

$$\Rightarrow \sum_i x_i = n\bar{x} = 200 \times 40 = 8000$$

Again, $\sigma^2 = \frac{1}{n}\sum_i x_i^2 - \left(\bar{x}\right)^2$

$$\Rightarrow \sum_i x_i^2 = n(\sigma^2 + \bar{x}^2) = 200 \times (225 + 1600) = 365000$$

$\therefore$ Corrected $\sum x_i = 8000 - 34 - 53 + 43 + 35 = 7991$

$\therefore$ Corrected $\sum x_i^2 = 365000 - 34^2 - 53^2 + 43^2 + 35^2 = 364109$

$\therefore$ Corrected mean $= \dfrac{7991}{200} = 39.95$ and

Corrected variance $= \dfrac{364109}{200} - (39.95)^2 = 224.14$

Example 5: An analysis of the monthly wages paid to the workers of two firms A and B belonging to the same industry gives the following results:

	Firm A	Firm B
No. of workers	500	600
Avg. monthly wages	186.00	175.00
Variance of wages	81	100

Calculate the average and standard deviation in both the firm together.

Solution: Let $\bar{x}$ and σ_x be the joint mean and standard deviation ,

Now, combined mean, $\bar{x} = \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \dfrac{500 \times 186 + 600 \times 175}{500 + 600} = \dfrac{198000}{1100} = 180$

Here, $d_1 = 186 - 180 = 6$ and $d_2 = 175 - 180 = -5$

Combined variance, $\sigma^2 = \dfrac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} = \dfrac{500(81+36) + 600(100+25)}{500+600} = 121.36$

Exercise 1: Consider the following data:

C.B.	f
15-25	4
25-35	11
35-45	19
45-55	14
55-65	0
65-75	2

Calculate Mean, Median, Mode, Standard Deviation, Q_3 and D_2.

Exercise 2: Find the suitable measure of skewness and kurtosis from the following data and determine the type of the distribution.

Class	f
0-20	20
20-50	50
50-100	64
100-250	30
250-500	25
500-1000	19

Exercise 3: The first one of two samples has 100 items with mean 15 and standard deviation 3. If the whole group has 250 items with mean 15.6 and standard deviation $\sqrt{13.44}$, then find standard deviation of second group.

Exercise 4: The sum and sum of squares corresponding to length X(in cm) and weight y(in gms) of 50 tapioca tubers are as follows:

$$\sum X = 212, \sum X^2 = 902.8, \sum y = 261, \sum y^2 = 1457.6$$

Which one is more varying in length and weight?